Volcano for the Agentic AI Era

Unified Scheduling Across Training, Inference, and Agents

KubeCon + CloudNativeCon North America 2026

SPEAKERS / AFFILIATIONS — TO CONFIRM

待确认：讲者姓名、署名顺序与单位；当前为工作版式，未套用官方模板。

Different Workloads, Shared Infrastructure

Different arrival patterns

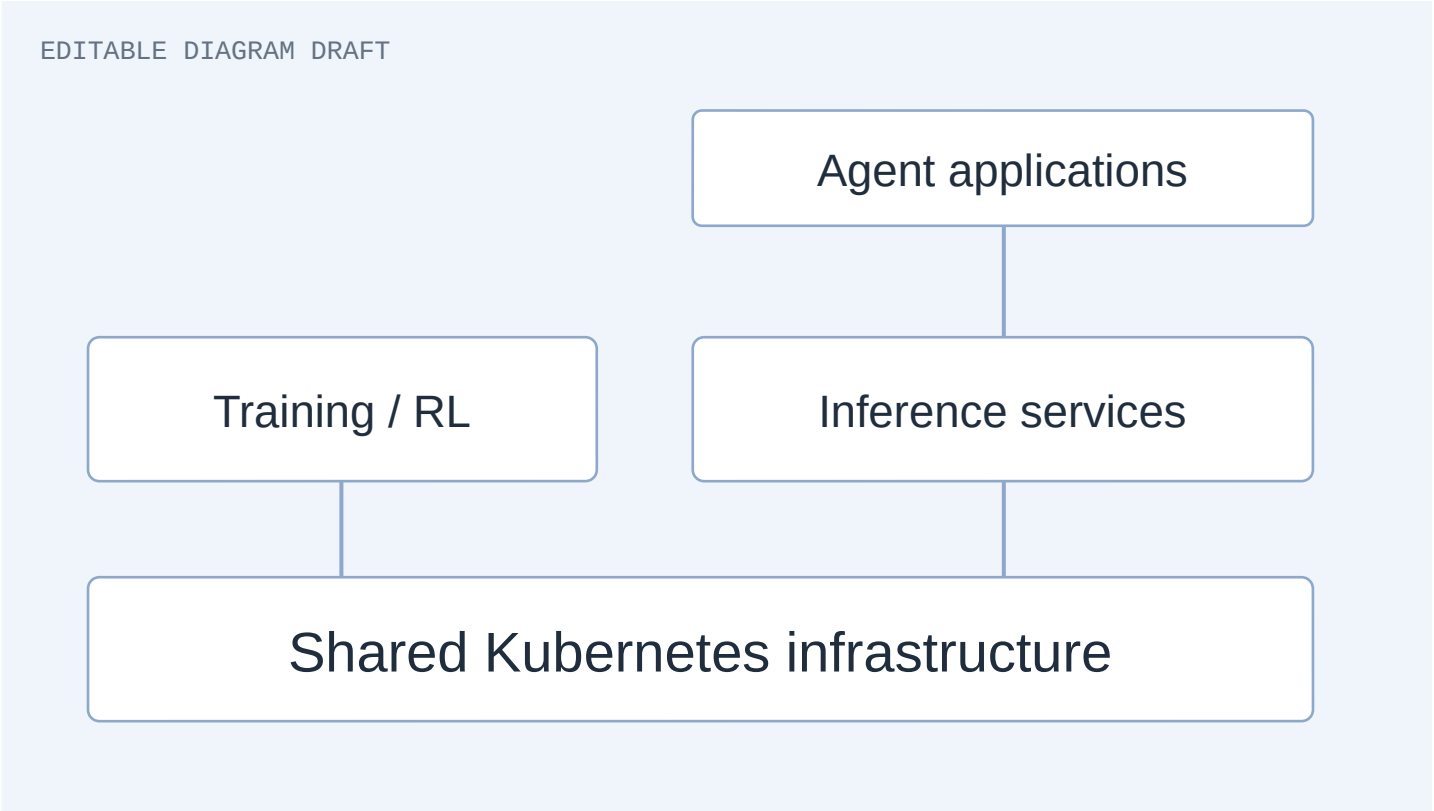
Long-running services, grouped compute, and bursts.

Different scheduling needs

Responsiveness, complete groups, and communication locality.

A shared resource pool

Coordinate infrastructure without forcing one execution model.



复用：生命周期图精简。 Agent 应用依赖推理服务，不把每次调用画成一个新 Pod 。

A Coordinated Scheduling Framework

Keep different scheduling paths

Support distinct workload requirements.

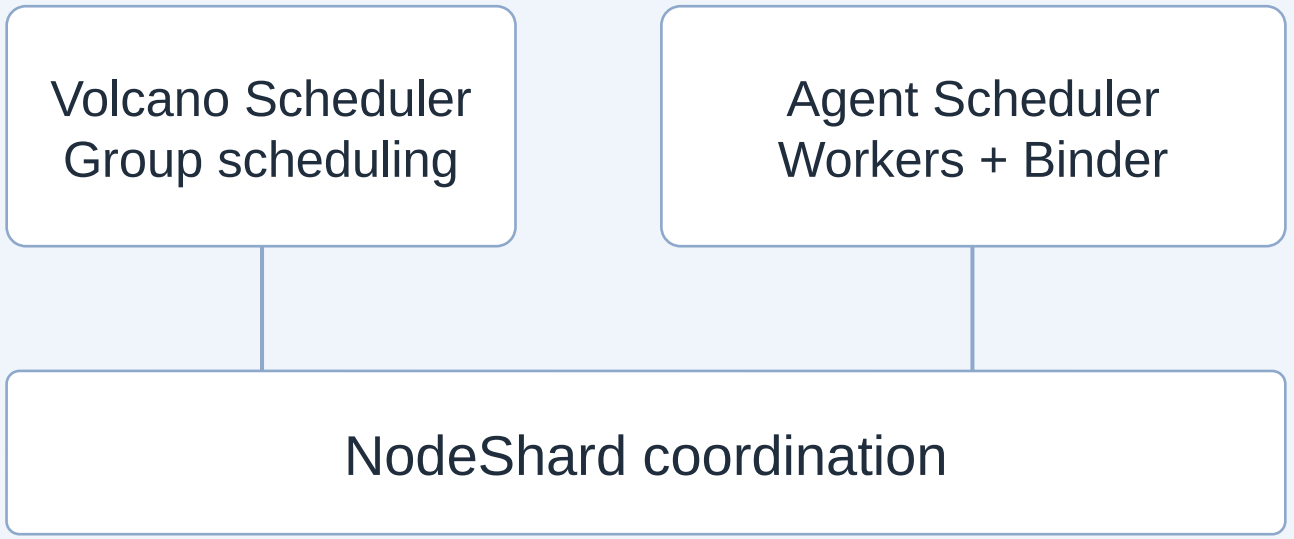
Coordinate access to nodes

NodeShard connects the paths at the infrastructure layer.

Handle local conflicts

Workers and Binder coordinate within the Agent Scheduler.

SCHEMATIC — REUSE / SIMPLIFY EXISTING ARCHITECTURE



Scheduler-internal conflicts ≠ cross-scheduler coordination

复用： China 7–8 / EU 9 。 worker / Binder 讲 30–45 秒；本页不放性能提升数字。

NodeShard: Coordinating Node Access

Controller sets desired membership

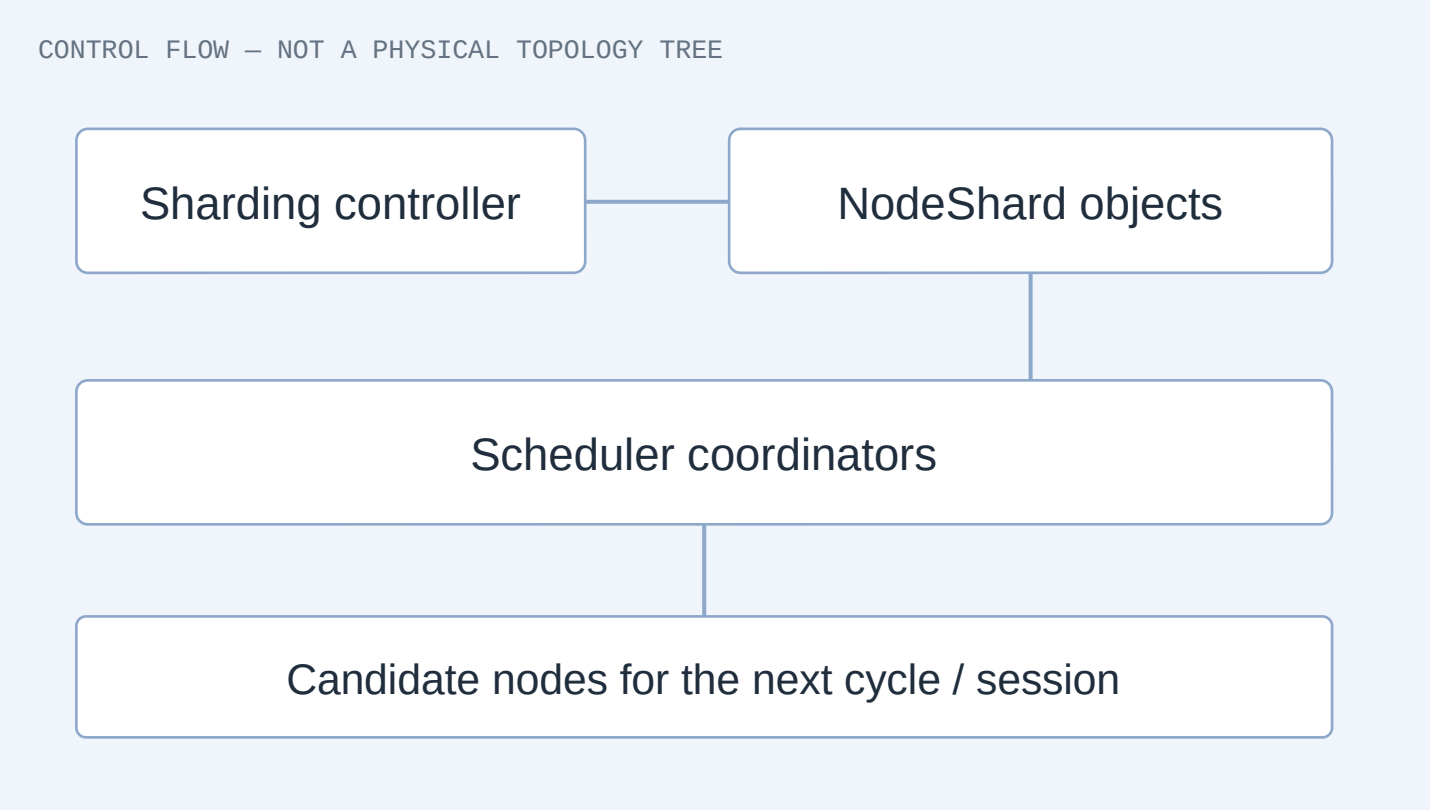
Express which nodes belong in each desired shard.

Coordinators synchronize usage

Reconcile desired and in-use nodes around scheduling cycles.

Mode defines the boundary

none: all nodes · soft: preference · hard: restricted scope



复用： China 9 / EU 10–11 。补一个节点范围交接例子；不是迁移运行中的 Pods 。

Available Nodes Are Only the Beginning

Enough free resources can still leave a workload unable to run.

01

Complete replicas

Do all required members fit?

02

Communication locality

Are the nodes in the right domain?

03

Group relationships

Should instances share a domain?

新增转场： NodeShard 是候选范围， HyperNode 是物理拓扑；不要画成父子层级。

PodGroup and SubGroup: Complete Replicas

PodGroup: overall gang requirements

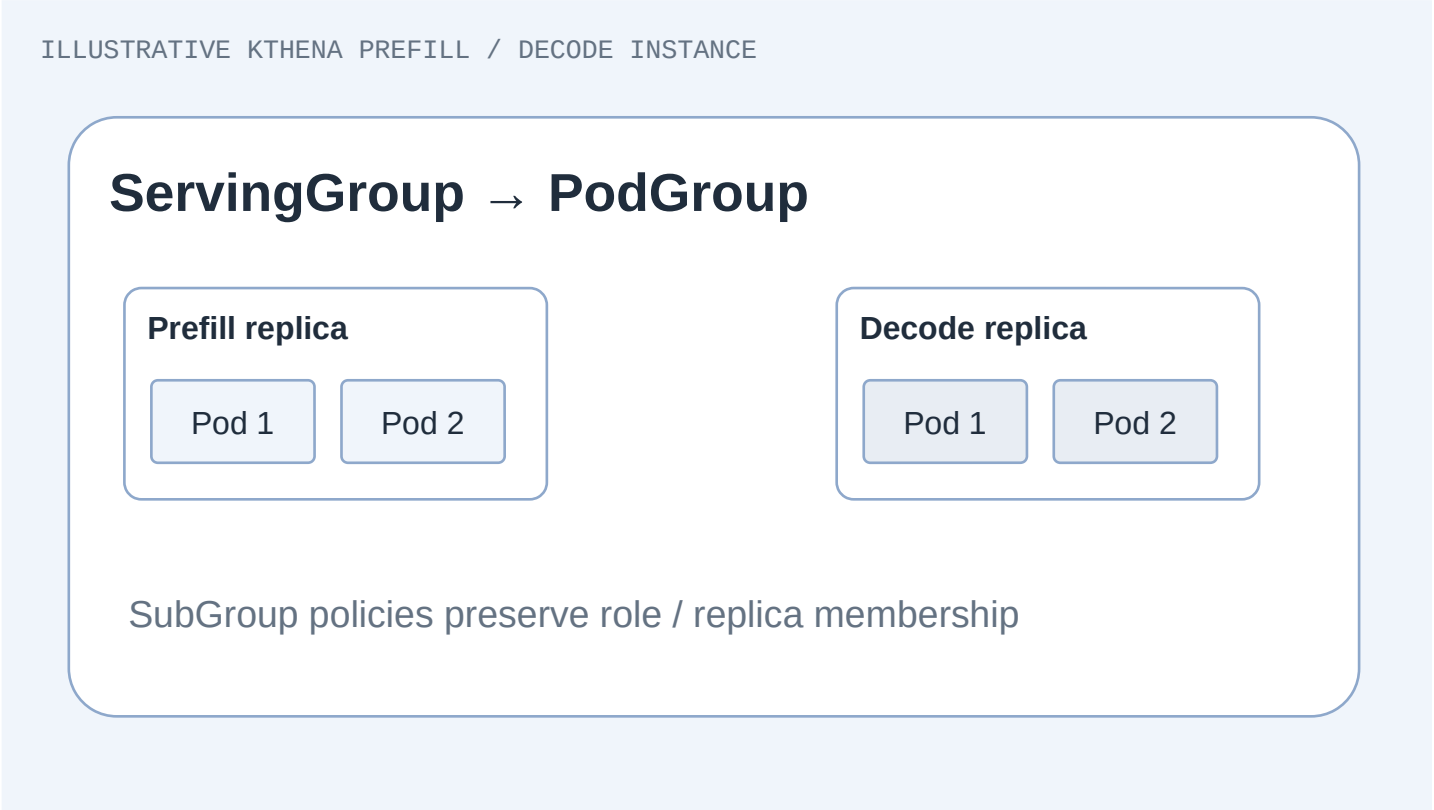
Describe what the workload needs as a whole.

SubGroup: complete member sets

Preserve a replica that spans multiple Pods.

Use role and replica identity

Extra Pods in one role cannot fill a missing replica in another.



复用： China 14 、 22 。 policy 可以产生多个实际 SubGroups ； 以下为 P/D 示意，不是完整 API 。

HyperNode: Expressing Topology Domains

Describe nested infrastructure domains

Represent nodes and their topology hierarchy.

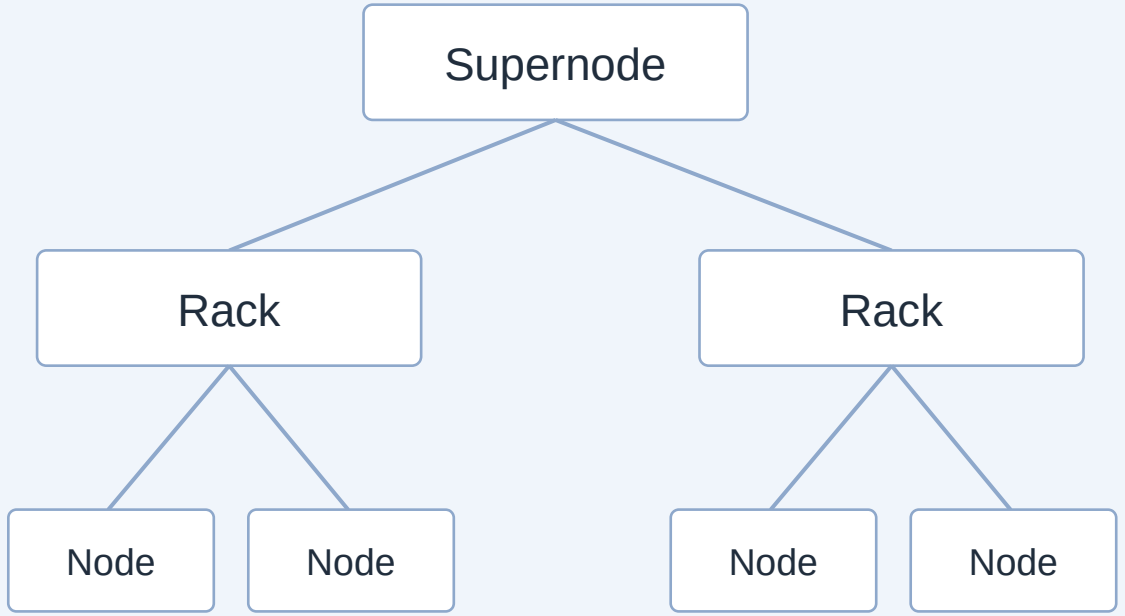
Choose a placement level

A group can target an appropriate domain size.

Express hard or soft constraints

State whether locality is required or preferred.

EDITABLE DIAGRAM DRAFT



复用： China 16 。仅保留一棵树；层级名称按现场集群映射，不写固定带宽收益。

Place Groups at Different Topology Levels

One boundary for the instance

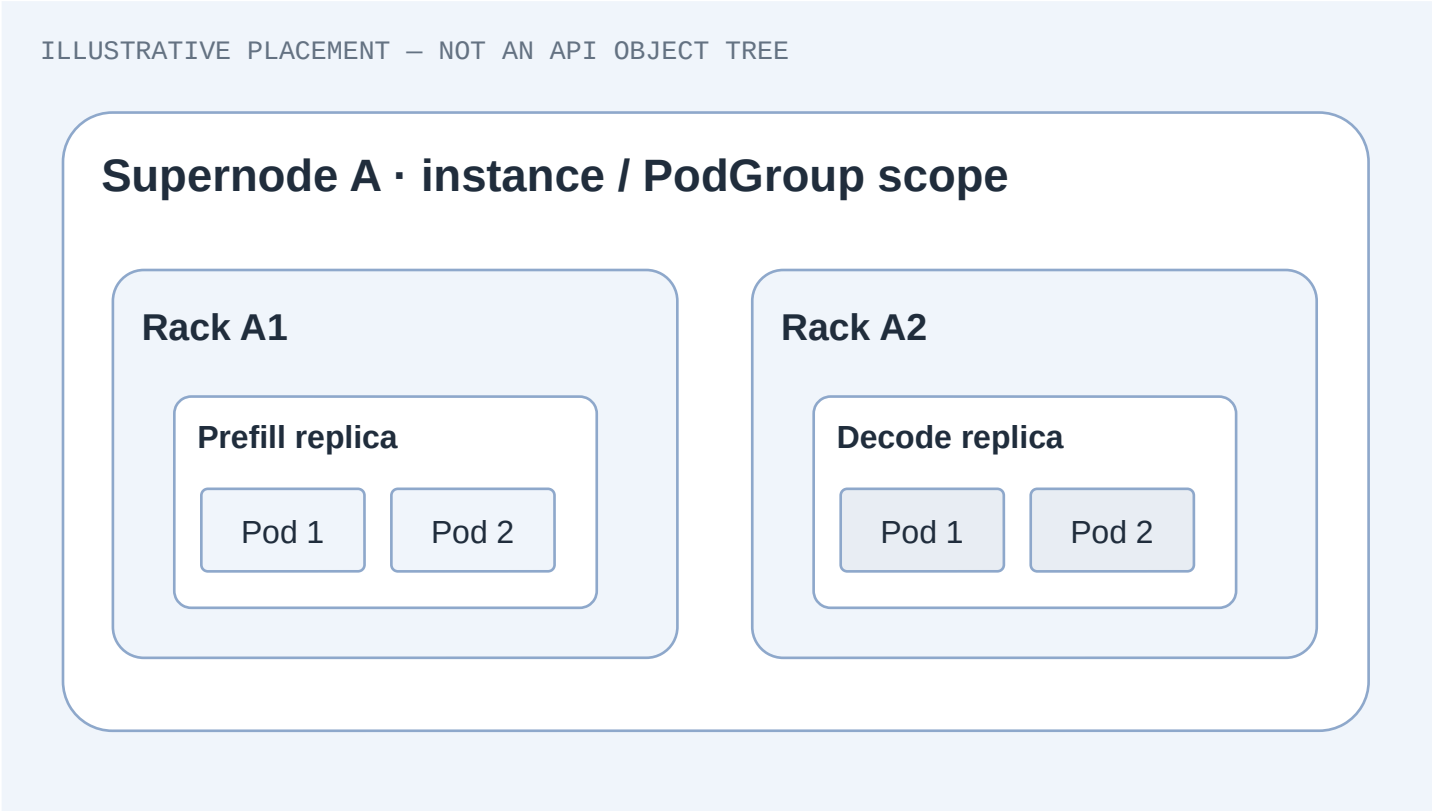
Keep the PodGroup inside a larger topology domain.

Tighter boundaries for replicas

Place each multi-Pod replica in a smaller domain.

Check capacity within the domain

Cluster-wide capacity alone is insufficient.



复用： China 17 。延续 P/D 案例；用两层边框区分实例范围和完整副本范围。

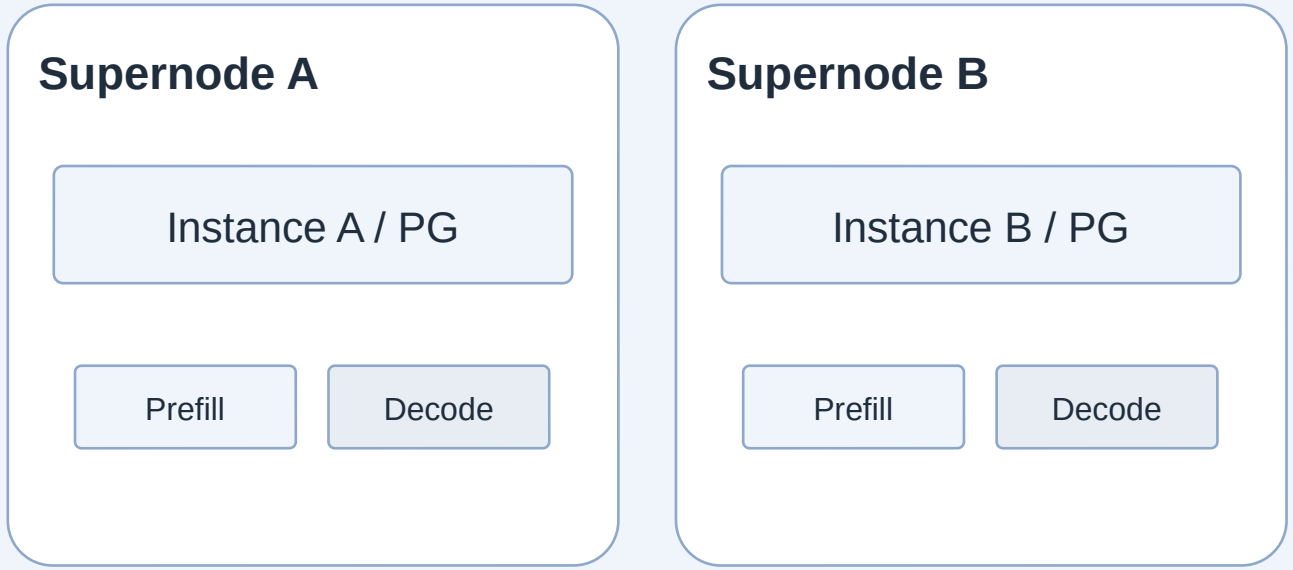
Keep Instances Together, Place Them Apart

AVAILABLE IN v1.16

Locality within.
Separation between.

Combine PodGroup network topology with anti-affinity across selected PodGroups.

EDITABLE DIAGRAM DRAFT



Within-group topology + between-group anti-affinity

SOURCE Volcano v1.16 · PodGroup anti-affinity #5349

新增重点： v1.16 PodGroup anti-affinity 。复用 China 15 的构图，更新功能状态。

When Placement Rules Cannot Be Satisfied

Example: only one domain has enough resources.

REQUIRED

Keep the second group Pending

Do not violate the separation rule.

PREFERRED

Allow a feasible fallback

The preference may be relaxed.

待补： 45–60 秒录屏或实际放置截图。当前是预期行为示意，不是测试结果。

Controllers Bring Workload Structure to Scheduling

PROJECT	EXISTING / MERGED	NEXT / IN PROGRESS
Kthena	Role / replica gang + topology	Reference use cases for group relationships
KubeRay	SubGroup support · #5249 merged	Replica topology · #5369 open
Grove	Gang / SubGroup backend merged	Topology mapping not yet supported
Kubeflow	Existing Volcano integration	Finer TrainJob groups + topology
LWS	Existing gang / Volcano integration	Explore roles / slices + group relationships

Controllers supply identity; the scheduler applies group and placement constraints.

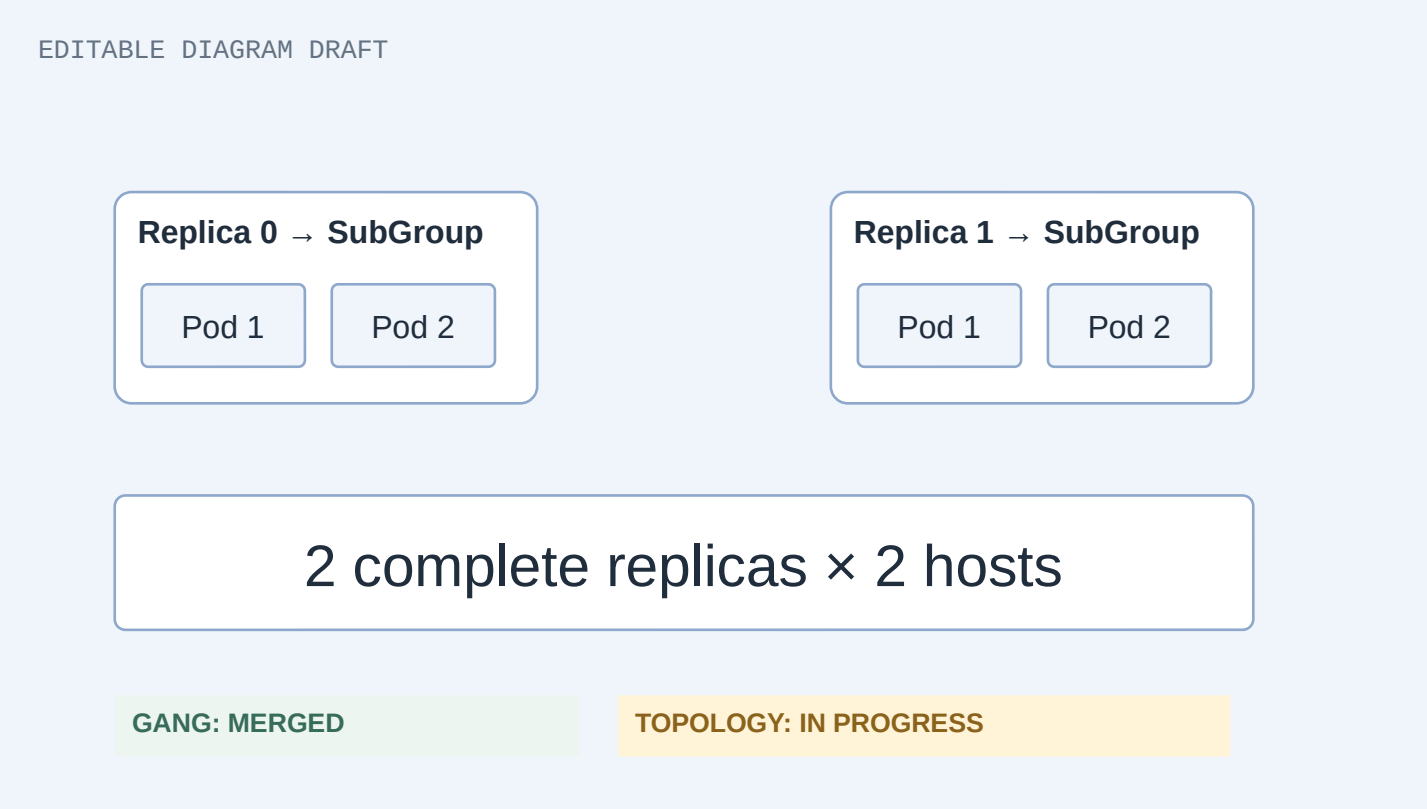
状态截至 2026-10-28；已合并不等于下游稳定版已发布。演讲前核对 PR 和版本。

KubeRay: A Replica Can Span Multiple Pods

```
minReplicas = 2
numOfHosts = 2
```

The requirement is two complete replicas, each spanning two Pods.

Any four Pods are not enough.



新增案例： WorkerGroup / policy / 实际 replica 不混为一谈；省略 head ，仅聚焦一个 worker group 。

HyperNode Beyond a Single Scheduler

Standalone discovery and controller

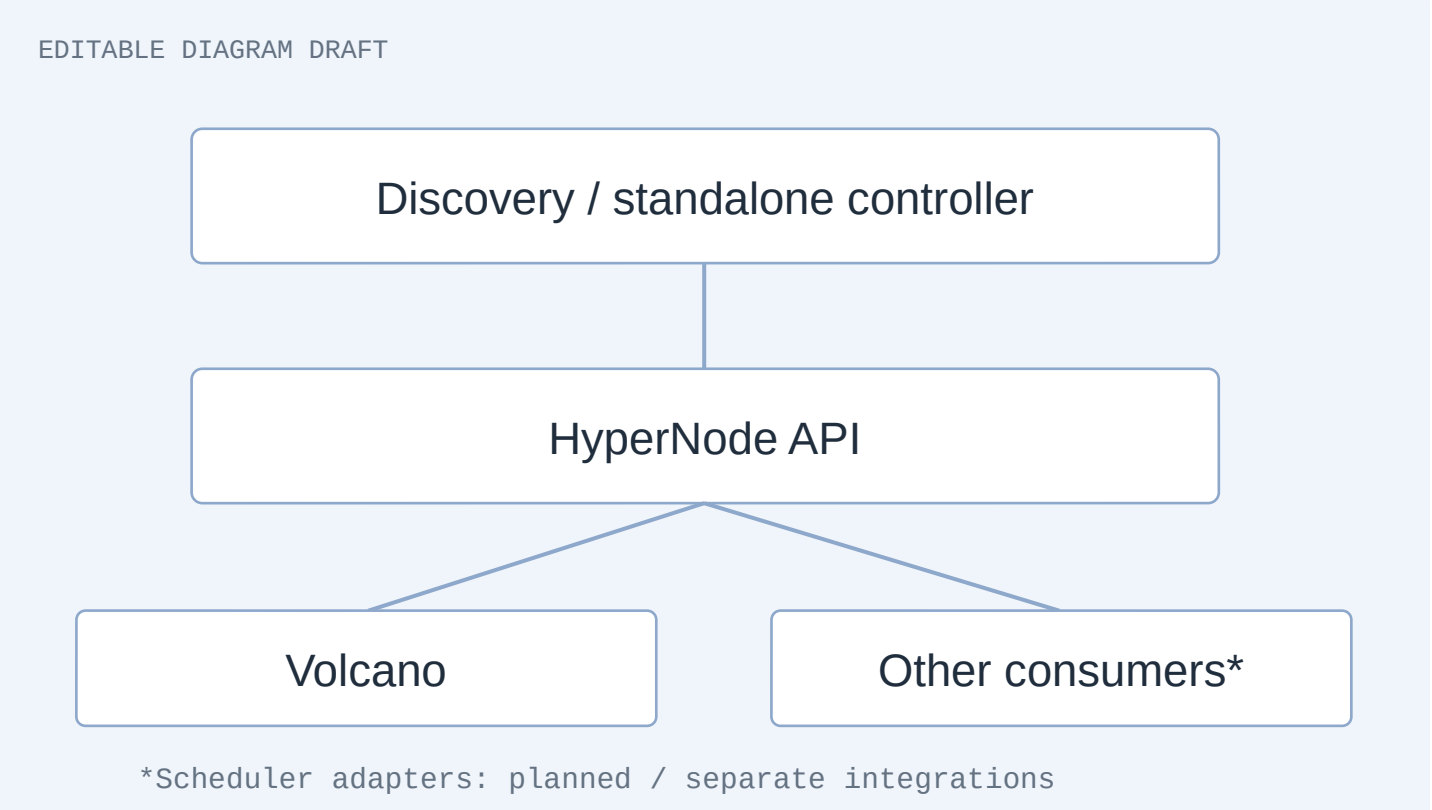
Topology can be maintained as a reusable component.

A shared topology API

Consumers read the HyperNode model.

Adapters remain separate work

API availability does not imply every scheduler is integrated.



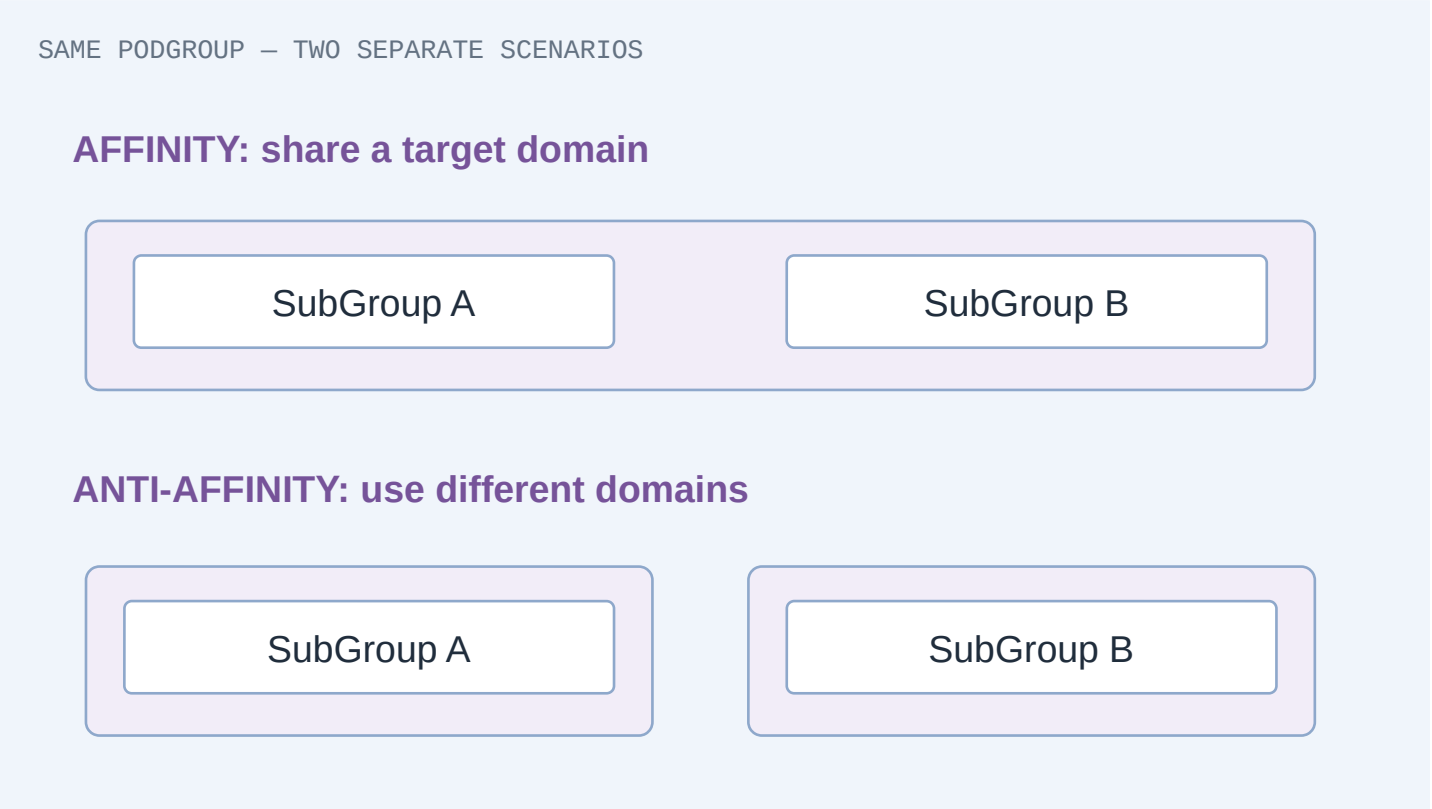
复用： China 19 。独立 controller 更新为已交付；外部 scheduler adapter 单独标状态。

Next: Relationships Between SubGroups

PROPOSED

Beyond a boundary for each group.

Express which SubGroups should be near each other, and which should be spread across domains.



未来方向：同一 PodGroup 内的 SubGroup 关系。以下只说明需求，不代表已发布字段。

From Placement Primitives to Ecosystem Adoption

01

Define relationships

SubGroup affinity / anti-affinity

Clear matching and failure semantics

02

Preserve workload intent

Deepen KubeRay / Grove mappings

Extend Kubeflow / LWS integration

03

Validate real scenarios

Runnable examples

Placement and failure behavior

Further out: topology adapters and device / network locality.

路线建议：与实际投入对齐；不承诺这页所有事项都在 v1.17 一次交付。

One Framework, Workload-Aware Placement

01

Coordinate scheduling paths

NodeShard and scheduler coordination

02

Preserve workload structure

Complete groups, topology, and placement relationships

03

Bring the semantics into controllers

Share use cases and help build integrations

github.com/volcano-sh/volcano · v1.17 roadmap: issue #6166

可用 20–30 秒提及 Global / 混部已有能力；正文收回到当前推进方向与社区参与。

Questions?

What should your workload keep together—or place apart?

Scheduling paths
NodeShard coordination

Workload groups
Gang + topology + relationships

github.com/volcano-sh/volcano

备份材料： NodeShard 模式、关键 API 、 plugin 开关、生态版本与来源；正式版移除此提示。